

Stone–Jordan Exceptional Theory

From the Void by Mirror Climb

Brandon Belna

June 6, 2026

Abstract

We formulate *Stone–Jordan Exceptional Theory* (SJET) from two primitive axioms: the **void** $\mathcal{V}$ (no observable distinction) and a **mirror operator** $\mathbf{M}$ that generates structure by involutive doubling. Iterating $\mathbf{M}$ on the Stone spectrum of $\mathcal{V}$ climbs the Hurwitz ladder $0 \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O} \rightarrow J_3(\mathbb{O})$ and the exceptional cascade $0 \rightarrow E_8 \rightarrow E_8 \times E_8 \rightarrow E_6 \rightarrow \cdots \rightarrow 0$; capacity balancing on the mirror-refined graph is the *derived* selection principle, not an independent axiom. At the E_6 rung we prove vacuum selection on EIII, composite $\text{Spin}(10) \times U(1)$ gauge theory with anomaly cancellation, and induced gravitation; postulates P1–P4 complete SM descent. The universe is the stabilized mirror climb $C^\infty(\mathcal{V})$, equivalently the capacity fixed point on the Albert–Cayley graph.

Keywords: void; mirror operator; Stone duality; exceptional Jordan algebra; E_6 coset; induced gravity.

MSC 2020: 03G05, 17C27, 22E46, 81T13.

Contents

1	Introduction	3
1.1	The climb picture	3
1.2	Epistemic conventions	3
1.3	Conventions	3
1.4	Organization	3
2	The Void and the Mirror Operator	3
2.1	The void	4
2.2	Category of Stone rungs	4
2.3	Rung table and the Hurwitz–Jordan theorems	4
2.4	The climb operator	5
3	The Mirror Climb: Division and Exceptional Ladders	5
3.1	Unified climb	6
3.2	Hurwitz rung	6
3.3	Exceptional rung	6
3.4	Capacity as dual mirror	7

4	Derived Arena: Graph, Albert Algebra, Order Parameter	7
4.1	Albert–Cayley graph	7
4.2	Stone lift	7
4.3	Order parameter (derived)	7
5	Dynamics: Capacity, Potential, and σ-Model	7
5.1	Derived potential	8
5.2	Vacuum selection	8
5.3	σ -model action	8
5.4	Universe as mirror fixed point	8
6	Emergent Spacetime and Time	8
6.1	Coset metric and irreducibility	9
6.2	Postulate P1: soldering	9
6.3	Emergent time	9
7	Gauge Sector and Induced Gravitation	9
7.1	Composite connection	9
7.2	Anomaly cancellation	10
7.3	Induced gravity	10
7.4	Renormalization group	10
8	Standard Model Descent	11
8.1	Postulates P2–P4	11
8.2	Descent chain	11
8.3	Scalar spectrum	11
9	Verification Ledger	11
9.1	Status table	11
9.2	Algebraic checks	11
9.3	Falsifiable predictions	12
10	Conclusions	12
10.1	Open problems	13
A	Albert Algebra and Cayley–Dickson Mirror	13
A.1	Cayley–Dickson mirror	13
A.2	Matrix model	13
A.3	Sharp map and rank	13
A.4	Freudenthal identity	13
	Acknowledgments	14
	References	14

1 Introduction

SJET asks: what is the minimal mathematical seed from which the Standard Model and gravitation can be *derived*, not merely embedded? Our answer is two-fold:

Axiom 1 The void $\mathcal{V}$ — no observable distinction, the terminal object of the Stone ladder.

Axiom 2 The mirror operator M — an involutive generator that, applied repeatedly, *climbs* the division and exceptional ladders.

Everything else — graphs, Albert algebra, capacity dual, E_6 potential, gauge sector — is **derived** from $M^n(\mathcal{V})$ and capacity balancing on mirror-refined partitions. Physical postulates P1–P4 (soldering, matter, families, Higgs) remain the honest boundary between proved mathematics and four-dimensional quantum field theory.

1.1 The climb picture

Schematically:

$$\mathcal{V} \xrightarrow{M} S_{\mathbb{R}} \xrightarrow{M} S_{\mathbb{C}} \xrightarrow{M} S_{\mathbb{H}} \xrightarrow{M} S_{\mathbb{O}} \xrightarrow{M} J_3(\mathbb{O}) \xrightarrow{M} \dots \xrightarrow{M} \mathcal{V}. \quad (1.1)$$

The bottom and top 0 are the same void: the climb is a *closed* ladder. At each rung, M acts as Boolean complement (discrete), Cayley–Dickson doubling (normed division), or heterotic/Cartan doubling (exceptional Lie). Capacity balancing $\mathcal{C}$ is the *dual mirror*: it selects the unique balanced partition at each level (BAMLP), not a separate axiom.

1.2 Epistemic conventions

- **Theorem** — proved from Axioms 1–2 and standard mathematics.
- **Derived** — explicit construction from mirror climb (no extra postulate).
- **Postulate** — added four-dimensional physics (P1–P4), named once.
- **Conjecture** — motivated but not controlled.

1.3 Conventions

Signature $(-, +, +, +)$; compact E_6 for dynamics; $\Phi \in J_3(\mathbb{O})$ order parameter; $\mathcal{C}$ capacity functional; M mirror operator; $\mathcal{V}$ void. Casimirs: $C_A(\text{Spin}(10)) = 8$, $T(\mathbf{16}) = 2$, $C_2(\mathbf{16}) = 45/8$. Branchings as in (3.2).

1.4 Organization

Section 2: void and mirror axioms; functor M ; Theorem 2.8 ($M^4(\mathcal{V}) = \mathbb{O}$, $M^5(\mathcal{V}) = J_3(\mathbb{O})$). Section 3: exceptional continuation and capacity dual. Section 4: derived graph, Stone lift, and order parameter. Section 5: capacity, potential, vacuum. Sections 6–8: physics under P1–P4. Section 9: ledger. Appendix A: Albert identities.

2 The Void and the Mirror Operator

This section states the *only* primitive axioms of SJET and gives the functorial definition of M on which all rung theorems depend.

2.1 The void

Definition 2.1 (Void). The *void* $\mathcal{V}$ is the trivial observable algebra: the two-element Boolean algebra $\mathcal{B}_0 = \{\perp, \top\}$ with $\perp < \top$, and its Stone spectrum $S_0 = \text{Spec}(\mathcal{B}_0)$, a single point. The associated division datum is the zero algebra $\{0\}$ of dimension 0. No vertex, no edge, no localisable quantum number exists at $\mathcal{V}$.

Axiom 2.2 (Void). Physical reality is anchored at the void $\mathcal{V}$: the state with no distinguishable observable. All structure arises by departure from $\mathcal{V}$, not by importing a graph or Lie group as primitive data.

2.2 Category of Stone rungs

Definition 2.3 (Stone rung category). Let **Stone** be the category of Stone spaces (compact, Hausdorff, totally disconnected) with continuous maps. A *rung datum* is a pair (S, D) with $S \in \mathbf{Stone}$ and D a formally real observable algebra at that rung. Morphisms preserve both S and the algebraic rung label.

Definition 2.4 (Three mirror modes). The mirror operator $\mathbf{M}$ acts in three modes, triggered by rung index:

M1. Discrete ($n = 0 \rightarrow 1$): append a Boolean generator a_{n+1} to $\mathcal{B}_n$, doubling clopen sectors: $\mathcal{B}_{n+1} = \mathcal{B}_n[a_{n+1}]/(a_{n+1}^2 = a_{n+1})$.

M2. Arithmetic ($1 \rightarrow 2 \rightarrow 3 \rightarrow 4$): Cayley–Dickson doubling $D_{n+1} = D_n \oplus D_n$ with product $(x, y) \cdot (u, v) = (xu - \bar{v}y, vx + y\bar{u})$ and involution $\overline{(x, y)} = (\bar{x}, -y)$ (Appendix A.1).

M3. Jordan exit ($4 \rightarrow 5$): because $\mathbf{M}_{\text{arith}}(\mathbb{O})$ is non-division (sedenions), the climb exits to $J_3(\mathbb{O}) = H_3(\mathbb{O})$, the unique formally real Jordan envelope of $\mathbb{O}$.

M4. Exceptional ($n \geq 6$): Lie mirror $G \mapsto G \times G$ or Cartan involution on G/H (Section 3.3).

Definition 2.5 (Mirror operator). A *mirror operator* is an endofunctor $\mathbf{M} : \mathbf{Stone} \rightarrow \mathbf{Stone}$ together with algebra maps such that:

- (i) $\mathbf{M}$ is **involutive** on stabilized rungs: $\mathbf{M}^2 \cong \text{id}$ (Boolean complement, field conjugation, coset exchange $\mathbf{16} \leftrightarrow \overline{\mathbf{16}}$).
- (ii) $\mathbf{M}$ is **nontrivial** at $\mathcal{V}$: $S_1 = \mathbf{M}(S_0)$ has $|S_1| = 2$ clopen points.
- (iii) $\mathbf{M}$ is **functorial**: compatible maps $f : S \rightarrow T$ induce $\mathbf{M}(f) : \mathbf{M}(S) \rightarrow \mathbf{M}(T)$.

Axiom 2.6 (Mirror). There exists $\mathbf{M}$ as in Definition 2.5 whose orbit from $\mathcal{V}$ generates the unified ladder (3.1). At each rung, capacity balancing $\text{Bal}_{\mathcal{C}}$ is the *dual mirror* (Theorem 3.10). The climb is closed: $\mathbf{M}^{\text{tot}}(\cdots (E_6) \cdots) = \mathcal{V}$.

2.3 Rung table and the Hurwitz–Jordan theorems

Definition 2.7 (Mirror power). Define rung algebras A_n and dimensions d_n by

$$(S_n, A_n) = \mathbf{M}^n(S_0, A_0), \quad (S_0, A_0) = (S(\mathcal{V}), \{0\}), \quad d_n = \dim_{\mathbb{R}} A_n. \quad (2.1)$$

Theorem 2.8 (Mirror rung table). *The first six arithmetic rungs of $\mathbf{M}^n(\mathcal{V})$ are:*

n	A_n	d_n	Stone points $ S_n $	Mirror mode
0	$\{0\}$	0	1	<i>void</i>
1	$\mathbb{R}$	1	2	<i>discrete</i>
2	$\mathbb{C}$	2	4	<i>arithmetic</i>
3	$\mathbb{H}$	4	8	<i>arithmetic</i>
4	$\mathbb{O}$	8	16	<i>arithmetic (Hurwitz terminal)</i>
5	$J_3(\mathbb{O})$	27	27	<i>Jordan exit</i>

In particular $\mathbf{M}^4(\mathcal{V})$ carries the octonions and $\mathbf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O})$.

Proof. Discrete step ($n = 0 \rightarrow 1$): adjoining one Boolean atom to $\mathcal{B}_0$ gives the four-element algebra $\{\perp, a, \neg a, \top\}$ with Stone spectrum two points; this is the first observable distinction, identified with ordered data $\mathbb{R}$ ($d_1 = 1$).

Arithmetic steps ($n = 1, 2, 3$): Cayley–Dickson doubling doubles dimension: $1 \rightarrow 2 \rightarrow 4 \rightarrow 8$. Hurwitz’s theorem (Theorem 3.3) proves no further *normed division* algebra exists at $d = 16$.

Jordan exit ($n = 4 \rightarrow 5$): $\mathbf{M}_{\text{arith}}(\mathbb{O})$ is the sedenion algebra $\mathbb{S}$ ($d = 16$), which fails division (Lemma 2.9). The climb therefore exits to the unique formally real Jordan algebra $H_3(\mathbb{O})$ with $d_5 = 27$ and cubic norm $N(X) = \det X$ (Proposition 3.4). The Stone spectrum S_5 has 27 clopen cells matching the 27 matrix coordinates. $\square$

Lemma 2.9 (Sedenion failure). *Cayley–Dickson doubling of $\mathbb{O}$ produces $\mathbb{S}$ of dimension 16 with zero divisors; $\mathbb{S}$ is not a division algebra. Hence $\mathbf{M}_{\text{arith}}^5(\mathcal{V}) \neq \mathbb{S}$ in the division category.*

Corollary 2.10 (Hurwitz terminal at $n = 4$). *$\mathbf{M}^4(\mathcal{V})$ is the last normed division rung. The next stable mirror power is $\mathbf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O})$, not a 16-dimensional division algebra.*

2.4 The climb operator

Definition 2.11 (Climb operator). The *climb operator* is

$$\mathbf{C} = \text{Bal}_{\mathcal{C}} \circ \mathbf{M}. \quad (2.2)$$

The *universe datum* is the profinite fixed point

$$\mathfrak{U}^* = \varprojlim_{n \rightarrow \infty} \mathbf{C}^n(\mathcal{V}). \quad (2.3)$$

At $n = 5$, $\mathbf{C}^5(\mathcal{V})$ stabilizes on the Albert–Cayley graph (Definition 4.1) with $1|10|16$ capacity partition.

Proposition 2.12 (Involution law). *On each stabilized rung, $\mathbf{M}^2 \cong \text{id}$ up to gauge: Boolean complement, field conjugation, and coset exchange $\mathbf{16}_{-3} \leftrightarrow \overline{\mathbf{16}}_{+3}$ are involutions.*

3 The Mirror Climb: Division and Exceptional Ladders

The ladders are the orbit of $\mathcal{V}$ under $\mathbf{M}$ (Axiom 2.6). Theorem 2.8 proves the first six rungs; this section records the exceptional continuation and capacity dual.

3.1 Unified climb

Definition 3.1 (Unified ladder). The *unified mirror climb* is

$$\mathcal{V} \xrightarrow{\mathbf{M}} \mathbb{R} \xrightarrow{\mathbf{M}} \mathbb{C} \xrightarrow{\mathbf{M}} \mathbb{H} \xrightarrow{\mathbf{M}} \mathbb{O} \xrightarrow{\mathbf{M}} J_3(\mathbb{O}) \xrightarrow{\mathbf{M}} E_8 \xrightarrow{\mathbf{M}} E_8 \times E_8 \xrightarrow{\mathbf{M}} E_6 \xrightarrow{\cdots} \mathcal{V}, \quad (3.1)$$

with n -th arrow $\mathbf{M}^n$ as in (2.1). Each arrow is followed by $\text{Bal}_{\mathcal{C}}$ where a partition is required.

Remark 3.2 (Rung index). $\mathbf{M}^4(\mathcal{V}) = \mathbb{O}$ and $\mathbf{M}^5(\mathcal{V}) = J_3(\mathbb{O})$ are **theorems** (Theorem 2.8), not definitions. The exceptional rungs $n \geq 6$ extend the Jordan stabilization.

3.2 Hurwitz rung

Theorem 3.3 (Hurwitz). *Mirror arithmetic doubling produces only*

$$\{0\}, \mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$$

of dimensions 0, 1, 2, 4, 8. Equivalently, $\mathbf{M}^4(\mathcal{V}) = \mathbb{O}$ and $\mathbf{M}_{\text{arith}}^k(\mathcal{V})$ is non-division for all $k \geq 5$ unless the Jordan exit to $J_3(\mathbb{O})$ is taken.

Proof. Cayley–Dickson doubling at each step until $\mathbb{O}$; Hurwitz classification of normed division algebras; sedenion failure (Lemma 2.9). $\square$

Proposition 3.4 (Terminal Jordan rung). *The mirror climb stabilizes at*

$$\mathbf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O}) = H_3(\mathbb{O}), \quad d_5 = 27.$$

The profinite limit $\varprojlim_n \mathbf{M}^n(\mathcal{V})$ has 27 observable coordinates. No 16-dimensional division rung appears.

Proof. From Theorem 2.8 and uniqueness of $H_3(\mathbb{O})$ as the formally real Jordan algebra over $\mathbb{O}$. $\square$

3.3 Exceptional rung

Proposition 3.5 (Exceptional mirror steps). *At $n \geq 6$ on the stabilized $J_3(\mathbb{O})$ rung, $\mathbf{M}$ acts as:*

- (i) E_8 : automorphism mirror of the $\mathbb{O}$ measure before projection;
- (ii) $E_8 \times E_8$: heterotic product mirror (left/right sectors);
- (iii) E_6 : rank-one capacity-balanced cell on $J_3(\mathbb{O})$;
- (iv) $E_6 \rightarrow H = \text{Spin}(10) \times U(1)$: Cartan mirror (EIII coset);
- (v) $\cdots \rightarrow \mathcal{V}$: collapse after electroweak breaking.

Proposition 3.6 (Branchings at E_6). *Under $H = \text{Spin}(10) \times U(1)$,*

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}, \quad (3.2)$$

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \quad (3.3)$$

$\mathcal{M} = E_6/H = \text{EIII}$, $\dim_{\mathbb{R}} \mathcal{M} = 32$, coset mirror $\mathbf{16}_{-3} \leftrightarrow \overline{\mathbf{16}}_{+3}$.

Conjecture 3.7 (Exceptional mirror from Jordan). *$\mathbf{M}^6(\mathcal{V}) = E_8$ and $\mathbf{M}^7(\mathcal{V}) = E_8 \times E_8$ are derivable from the $G_2 \subset F_4 \subset E_6$ stabilizer tower on $J_3(\mathbb{O})$ without importing heterotic string theory as primitive data.*

Remark 3.8 (Heterotic parallel). $E_8 \times E_8$ matches the heterotic gauge group; $h^1(\hat{Q}, V_{16}) = 3$ [11] corroborates three generations (Remark 8.9).

3.4 Capacity as dual mirror

Definition 3.9 (Capacity functional). On the mirror-refined Albert graph $\mathcal{G}$ with costs c_i and μ ,

$$\mathcal{C}(\omega) = - \sum_{a \in V} \mu(a) \log \left(\sum_i e^{c_i(a) - \omega_i} \right). \quad (3.4)$$

$\mathcal{C}$ is the *dual mirror*: $\mathbf{M}$ doubles sectors; $\text{Bal}_{\mathcal{C}}$ balances them.

Theorem 3.10 (Concavity and dual existence). $\mathcal{C}$ is strictly concave on $\{\omega : \sum_i \omega_i = 0\}$. For targets $m_i > 0$, $\sum_i m_i = 1$, there exists unique ω^* with $\partial \mathcal{C} / \partial \omega_i(\omega^*) = m_i$.

Theorem 3.11 (Capacity at the Jordan rung). On $\mathcal{G}_{\text{Albert}}$ at $\mathbf{M}^5(\mathcal{V})$, the unique capacity maximizer yields the 1|10|16 partition of Proposition 4.2. This is the mirror-dual of the **27** branching (3.2).

4 Derived Arena: Graph, Albert Algebra, Order Parameter

This section contains no new axioms. The Albert–Cayley graph, Stone lift, and order parameter are **derived** from $\mathbf{C}^\infty(\mathcal{V})$.

4.1 Albert–Cayley graph

Definition 4.1 (Albert–Cayley graph). At the $J_3(\mathbb{O})$ rung of (3.1), the mirror-refined graph $\mathcal{G}_{\text{Albert}} = (V, E, w)$ has $V = \{1, \dots, 27\}$ (basis of $J_3(\mathbb{O})$), edges where $\partial_a \partial_b \text{Tr}(X^2) \neq 0$, uniform μ , and territory generators z_1, z_{10}, z_{16} for the sectors of (3.2).

Proposition 4.2 (BAMLP partition). Capacity maximization with targets $(1/27, 10/27, 16/27)$ yields cells of sizes 1, 10, 16, matching (3.2).

4.2 Stone lift

Theorem 4.3 (Profinite Stone limit). Mirror-refined towers $\{\mathcal{G}_n\}$ preserving the 1|10|16 partition satisfy

$$\text{Lift}(\mathcal{G}_\infty) \cong J_3(\mathbb{O}), \quad S_\infty = \varprojlim_n \text{Spec}(\mathcal{B}(\mathcal{G}_n)).$$

4.3 Order parameter (derived)

Definition 4.4 (Order parameter). The *order parameter* is the smooth section $\Phi : M^4 \rightarrow J_3(\mathbb{O})$ transforming in **27** of compact E_6 . It is the continuum limit of coordinates on $\text{Lift}(\mathcal{G}_\infty)$, not an independent axiom.

Theorem 4.5 (Invariants and orbits). Only $N(X) = \det X$ is E_6 -invariant (up to character); $Q(X) = \text{Tr}(X^2)$ is F_4 -invariant only. Real orbits are classified by $\text{rank } X$ and $N(X)$; $\text{rank } X \leq 1 \iff X^\# = 0$ (Appendix A.3).

Proposition 4.6 (Vacuum manifold). The breaking orbit is $\mathcal{M} = E_6/H = \text{EIII}$, $\dim_{\mathbb{R}} \mathcal{M} = 32$, with coset tangent $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$.

5 Dynamics: Capacity, Potential, and σ -Model

Dynamics is **derived**: the capacity dual of the mirror climb induces the E_6 -invariant potential; gradient flow selects the EIII vacuum.

5.1 Derived potential

Definition 5.1 (Continuum potential). The SJET potential on $J_3(\mathbb{O})$ is

$$V(X) = \alpha \langle X^\#, X^\# \rangle + \beta [\text{Tr}(X^2) - c_2]^2 + \gamma \text{Tr}(X^2)^k, \quad \alpha, \beta, \gamma > 0, \quad k \geq 2 \text{ even}, \quad (5.1)$$

the profinite limit of the graph penalty under $\text{Bal}_{\mathcal{C}}$ on $\mathcal{G}_{\text{Albert}}$.

Theorem 5.2 (Potential identification). *V is the unique E_6 -invariant potential with quartic $\langle X^\#, X^\# \rangle$ vanishing on $\text{rank} \leq 1$, induced by the mirror-dual capacity on $\mathcal{G}_{\text{Albert}}$.*

5.2 Vacuum selection

Lemma 5.3 (Sharp locus). $\langle X^\#, X^\# \rangle \geq 0$ with equality iff $\text{rank } X \leq 1$.

Theorem 5.4 (Global minimum on EIII). *V is coercive on compact E_6 . Global minima lie on the rank-one orbit $\mathcal{M} = \text{EIII}$; the origin is a strict local maximum.*

Corollary 5.5 (Tree-level Goldstones). $\text{Hess } V(\Phi_\star) = 0$ on $\mathfrak{m}$: 32 exact Goldstones at tree level.

5.3 σ -model action

$$S[\Phi] = \int d^4x \sqrt{|g|} \left[\frac{1}{2} K_{AB} \nabla_\mu \Phi^A \nabla^\mu \Phi^B + V(\Phi) \right], \quad (5.2)$$

with K the unique H -invariant metric on $\mathfrak{m}$ (Theorem 6.2).

5.4 Universe as mirror fixed point

Theorem 5.6 (Universe as $C^\infty(\mathcal{V})$). *Under Axioms 2.2–2.6 and Postulates P1–P4, the universe datum $\mathfrak{U}^\star$ of (2.3) satisfies:*

- (i) $\omega^\star$ maximizes $\mathcal{C}$ on $\mathcal{G}_{\text{Albert}}$;
- (ii) $\Phi_\star \in \mathcal{M}$ minimizes V ;
- (iii) $g_{\mu\nu}$ is the soldered Minkowski metric (P1);
- (iv) $\mathcal{H}$ is the three-family SM (P2–P4).

Proof. (i)–(ii) from Theorems 3.10 and 5.4. (iii)–(iv) from postulates. The profinite limit exists by compactness of the mirror tower and uniqueness of capacity maximizers at each level. $\square$

6 Emergent Spacetime and Time

Four-dimensional Lorentzian geometry is not a consequence of the E_6 coset metric alone. Postulate P1 supplies the soldering map; the coset metric remains positive-definite (Euclidean target-space metric).

6.1 Coset metric and irreducibility

Theorem 6.1 (Real-irreducibility of $\mathfrak{m}$). *As a real H -module, $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ is irreducible of complex type: $\text{End}_H(\mathfrak{m}) \cong \mathbb{C}$.*

Proof. Complexifying, $\mathfrak{m}_{\mathbb{C}} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ with non-isomorphic summands exchanged by conjugation. A real submodule would be a complex submodule invariant under conjugation; Schur's lemma on each summand forbids nontrivial proper submodules. $\square$

Theorem 6.2 (Unique coset metric). *The restriction $K|_{\mathfrak{m}}$ of the Killing form is the unique H -invariant positive-definite inner product on $\mathfrak{m}$.*

Proof. H -invariant bilinear forms correspond to H -invariant maps $\mathfrak{m} \otimes \mathfrak{m} \rightarrow \mathbb{R}$. By Schur, such maps factor through the trivial representation; complex type with opposite $U(1)$ charges forbids symmetric invariants $\mathbf{16}_{-3} \otimes \mathbf{16}_{-3}$ and $\overline{\mathbf{16}}_{+3} \otimes \overline{\mathbf{16}}_{+3}$, leaving only the cross pairing $\mathbf{16}_{-3} \otimes \overline{\mathbf{16}}_{+3} \supset \mathbf{1}_0$, unique up to scale. $\square$

Remark 6.3. $K|_{\mathfrak{m}}$ is **positive-definite**, not Lorentzian. Signature $(-, +, +, +)$ is carried by the soldering form, not by the Killing restriction.

6.2 Postulate P1: soldering

The 2×2 Hermitian-octonionic block embeds as $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ with $\det$ the Minkowski norm; $\text{SL}(2, \mathbb{O}) \cong \text{Spin}(1, 9)$.

Postulate 6.4 (Soldering and dimension, P1). *There exists a soldering map σ identifying four clock-field directions e_{μ}^a ($a = 0, 1, 2, 3$) with a four-dimensional subspace of $H_2(\mathbb{O})$ carrying $\eta_{ab} = \text{diag}(-1, +1, +1, +1)$. The world manifold has $d = 4$. The vierbein emerges from the Maurer–Cartan $\mathfrak{m}$ -component (Section 7).*

Proposition 6.5 (Emergent vierbein). *For a coset section $g(x) \in E_6$ with $g^{-1}\partial_{\mu}g = A_{\mu} + e_{\mu}$, the $\mathfrak{m}$ -component defines a composite vierbein. Under P1, $\eta_{\mu\nu} = e_{\mu}^a e_{\nu}^b \eta_{ab}$ has signature $(-, +, +, +)$.*

6.3 Emergent time

Proposition 6.6 (Page–Wootters clock). *In the semiclassical regime of induced gravity (Theorem 7.4), the Hamiltonian constraint $H_{\text{phys}}|\psi\rangle = 0$ holds on physical states. The soldered timelike clock Φ^0 defines internal time; conditional evolution $U(\tau) = e^{-iH_{\tau}\tau}$ on physical states recovers standard quantum dynamics.*

Remark 6.7 (OS reconstruction). Euclidean correlators follow from (5.2). Lorentzian unitary dynamics requires Osterwalder–Schrader reconstruction along Φ^0 . **Conjecture:** the σ -model satisfies reflection positivity.

7 Gauge Sector and Induced Gravitation

7.1 Composite connection

The Lie algebra splits H -equivariantly as $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}_{10} \oplus \mathfrak{u}_1$ and $\mathfrak{m}$ as in (3.3).

Theorem 7.1 (Composite gauge connection). *For $g(x) \in E_6$, write $g^{-1}\partial_\mu g = \mathcal{A}_\mu + e_\mu$ with $\mathcal{A}_\mu \in \mathfrak{h}$, $e_\mu \in \mathfrak{m}$. Under $g \mapsto gh^{-1}$, $h \in H$,*

$$\mathcal{A}_\mu \mapsto h\mathcal{A}_\mu h^{-1} - (\partial_\mu h)h^{-1},$$

and the curvature $\mathcal{F}_{\mu\nu} = \partial_{[\mu}\mathcal{A}_{\nu]} + [\mathcal{A}_\mu, \mathcal{A}_\nu]$ transforms as an H field strength. All 46 generators of $\mathfrak{h}$ remain massless because H is unbroken.

Proof. Standard transformation of the canonical connection on the reductive homogeneous bundle $E_6 \rightarrow \mathcal{M}$. $\square$

Proposition 7.2 (Induced Yang–Mills). *At one loop, integrating out massive coset modes generates $S_{\text{YM}} = \frac{1}{4\hat{g}^2} \int \text{Tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})$ with $\hat{g}^{-2} \propto f^2\mu^2/(16\pi^2)$, $[f] = \text{mass}$.*

7.2 Anomaly cancellation

Anomalies are computed with Dynkin indices $A_{\mathbf{16}} = A_{\mathbf{10}} = 1$, $A_{\mathbf{1}} = 0$ and $U(1)$ charges from (3.2).

Theorem 7.3 (Green–Schwarz cancellation). *The cubic anomaly $\mathcal{A}_{U(1)^3}$ vanishes: $16 \cdot 1^3 + 10 \cdot (-2)^3 + 1 \cdot 4^3 = 0$. The mixed anomaly $\mathcal{A}_{U(1)\cdot\text{Spin}(10)^2}$ has coefficient -1 : $1 \cdot 1 + (-2) \cdot 1 + 4 \cdot 0 = -1$, and is cancelled by a Green–Schwarz term $\propto \text{tr}(F \wedge F) B$ with axion B from the $\mathbf{1}_{+4}$ sector.*

Proof. Direct arithmetic on charges and indices; see [6] for the mechanism. $\square$

7.3 Induced gravity

Theorem 7.4 (Sakharov induced gravity). *In a heat-kernel scheme, only massive modes contribute to the a_2 coefficient. With 28 physical massive coset scalars ($s = +1$, $m^2 = \mu^2$ canonically),*

$$M_{\text{Pl}}^2 = \frac{1}{16\pi^2} \cdot \frac{1}{6} \sum_{\text{massive}} s_i m_i^2 \log \frac{\Lambda^2}{m_i^2} \approx \frac{28}{6} \cdot \frac{\mu^2}{16\pi^2} \log \frac{\Lambda^2}{\mu^2} > 0. \quad (7.1)$$

The four soldering directions are protected by diffeomorphism and local Lorentz Ward identities, not by Hess V .

Remark 7.5 (Cosmological constant). The a_0 term yields an uncanceled Λ^4 vacuum energy. SJET does not solve the cosmological constant problem; Λ_{cc} is a tuned relevant coupling.

7.4 Renormalization group

Theorem 7.6 (Gauge β -function). *With canonical dimensionless coupling $\hat{g}^2$,*

$$\beta_{\hat{g}^2} = 2\hat{g}^2 + \frac{b_0}{48\pi^2}\hat{g}^4, \quad b_0 = \frac{11}{3}C_A - \frac{4}{3}T n_F - \frac{1}{6}T n_S. \quad (7.2)$$

For three chiral $\mathbf{16}s$ ($n_F = 3$) and $n_S = 28$ coset scalars:

$$b_0(3 \text{ gen}) = \frac{11}{3}(8) - \frac{4}{3}(2)(3) - \frac{1}{6}(2)(28) = \frac{88 - 24 - 28}{3} = 12. \quad (7.3)$$

The one-generation anchor ($n_F = 1$, $n_S = 2$) gives $b_0(1 \text{ gen}) = 26$. The gauge sector is asymptotically free ($\hat{g}_\star^2 = 0$).

Conjecture 7.7 (Gravitational UV fixed point). *A truncated RG gives $\beta_\kappa = 2\kappa - (5/48\pi^2)\kappa^2$ with UV-attractive $\kappa_\star = 96\pi^2/5$. This one-loop result is indicative, not rigorous [9].*

8 Standard Model Descent

Matter and electroweak symmetry breaking are **postulated** (P2–P4). Anomaly cancellation is a **theorem** once the **27** content is fixed.

8.1 Postulates P2–P4

Postulate 8.1 (Chiral matter, P2). Each family furnishes one chiral **16** of $\text{Spin}(10)$ with charges from (3.2). No mirror $\overline{\mathbf{16}}$ partners.

Postulate 8.2 (Three families, P3). Horizontal $SU(3)_F$ acts on $J_3(\mathbb{O}) \otimes \mathbb{C}^3$; the three factors are three chiral **16**s. Three generations are *not* derived from Axioms I–III.

Postulate 8.3 (GUT–Higgs and Yukawa, P4). Breaking proceeds via $\mathbf{45}_H$ or $\mathbf{54}_H$ ($\text{Spin}(10) \rightarrow SU(5) \times U(1)_X$), $\mathbf{126}_H$ (breaks $B - L$), and $\mathbf{10}_H$ (electroweak). Yukawas $y_{ij} \mathbf{16}_i \mathbf{16}_j H$ generate fermion masses; type-I seesaw from $\mathbf{126}_H$ gives $m_\nu \sim m_D^2/M_R$.

Postulate 8.4 (Flavon hierarchy, P3'). Sequential $SU(3)_F$ breaking by flavon VEVs generates Yukawa hierarchy and CKM/PMNS mixing.

8.2 Descent chain

Theorem 8.5 (SM gauge structure under P4). *Assume P4. The breaking chain*

$$E_6 \xrightarrow{V} \text{Spin}(10) \times U(1) \xrightarrow{\mathbf{45}_H} SU(5) \times U(1)_X \xrightarrow{\mathbf{126}_H} SU(3)_c \times SU(2)_L \times U(1)_Y \xrightarrow{\mathbf{10}_H} SU(3)_c \times U(1)_{\text{em}}$$

yields the SM gauge group with GUT normalization $\alpha_1 = (5/3)\alpha_Y$.

Per family $\mathbf{16} = (q, u^c, d^c, \ell, e^c, \nu^c)$; $\mathbf{10}_H$ supplies $v = 246 \text{ GeV}$.

Remark 8.6 (Yukawa source term). A bare coupling $\bar{\psi} \Phi^a T_a \psi$ with Φ in $\{\mathbf{16}, \overline{\mathbf{16}}\}$ is *not* H -invariant and does not generate masses at the $\Phi_\star$ vacuum. Masses require P4 Higgs multiplets [7].

8.3 Scalar spectrum

Theorem 8.7 (Tree-level Goldstones). *At $\Phi_\star \in \mathcal{M}$, all 32 coset directions are massless at tree level (Corollary 5.5).*

Theorem 8.8 (Radiative degeneracy). *The Coleman–Weinberg potential [10] is H -invariant. Since $\mathfrak{m}$ is a single real-irreducible H -module (Theorem 6.1), Schur gives one eigenvalue on all 28 physical massive scalars; canonical mass $m^2 = \mu^2$.*

Remark 8.9 (Heterotic corroboration). The heterotic $\mathbb{Z}_5$ quintic gives $h^1(\hat{Q}, V_{16}) = 3$ [11]. **Conjecture**, not a derivation of P3.

9 Verification Ledger

9.1 Status table

9.2 Algebraic checks

Anomalies: $16 - 80 + 64 = 0$; mixed $1 - 2 + 0 = -1$. β : $b_0 = 12$. Dimensions: $78 = 45 + 1 + 16 + 16$; $27 = 1 + 10 + 16$; $32 = 28 + 4$.

Claim	Status	Check	Ref.
Void $\mathcal{V}$ primitive	Axiom	$\mathcal{B}_0 = \{\perp, \top\}$	2.2
Mirror $\mathbf{M}$ primitive	Axiom	functor on Stone	2.6
$\mathbf{M}^4(\mathcal{V}) = \mathbb{O}$	Theorem	Hurwitz terminal	2.8
$\mathbf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O})$	Theorem	Jordan exit	2.8
Sedenion failure	Lemma	zero divisors	2.9
Hurwitz ladder	Theorem	dims 0, 1, 2, 4, 8	3.3
$\mathbf{C}^\infty(\mathcal{V}) \cong J_3(\mathbb{O})$	Theorem	27 stable	4.3
1 10 16 partition	Prop.	capacity	4.2
Potential from $\mathcal{C}$	Theorem	limit	5.2
EIII vacuum	Theorem	coercivity	5.4
Coset mirror $16 \leftrightarrow \overline{16}$	Theorem	$U(1)$ charge	3.6
Anomaly cancellation	Theorem	$16 - 80 + 64$; GS -1	7.3
$b_0(3) = 12$	Theorem	(7.3)	7.6
$\mathfrak{U}^* = \mathbf{C}^\infty(\mathcal{V})$	Theorem	under P1–P4	5.6
Soldering, $d = 4$	Postulate	P1	6.4
Matter, 3 families	Postulate	P2–P3	§8.1
GUT–Higgs	Postulate	P4	8.3
$\kappa_\star$	Conjecture	1-loop	7.7
Heterotic $h^1 = 3$	Conjecture	string	8.9

Table 1: Epistemic ledger. Only two axioms: void and mirror.

9.3 Falsifiable predictions

F1. Proton decay bound on M_{GUT} .

F2. Seesaw neutrino scale $m_\nu \sim 0.05 \text{ eV}$.

F3. No fourth generation (mirror climb stabilizes at three $SU(3)_F$ copies).

F4. Common CW mass μ on 28 coset scalars.

F5. Fourth generation would break $b_0 > 0$ window.

10 Conclusions

SJET reduces its primitive content to **two axioms**:

Axiom 1 the **void** $\mathcal{V}$ — no observable distinction;

Axiom 2 the **mirror operator** $\mathbf{M}$ — involutive doubling that climbs the division and exceptional ladders.

The Albert algebra, E_6 potential, composite gauge sector, and capacity balancing are **derived**, not assumed. The universe is

$$\mathfrak{U}^* = \mathbf{C}^\infty(\mathcal{V}) = \varprojlim_n (\text{Bal}_{\mathcal{C}} \circ \mathbf{M})^n(\mathcal{V}), \quad (10.1)$$

stabilizing at $J_3(\mathbb{O})$ and E_6 , with physical contact via P1–P4.

10.1 Open problems

- O1.** Prove Conjecture 3.7: $M^6(\mathcal{V}) = E_8$, $M^7(\mathcal{V}) = E_8 \times E_8$ from $J_3(\mathbb{O})$ alone.
- O2.** Derive P3 (three families) from mirror stabilisation at M^7 .
- O3.** Full categorical equivalence $\mathbf{Stone} \simeq \mathbf{Div}^{\text{norm}}$ along the climb.
- O4.** Gravitational UV fixed point beyond one-loop truncation.
- O5.** Cosmological constant.

A Albert Algebra and Cayley–Dickson Mirror

A.1 Cayley–Dickson mirror

Definition A.1 (Cayley–Dickson doubling). Given a normed algebra $(D, \bar{\cdot}, |\cdot|)$, the *mirror double* is $D^M = D \oplus D$ with product

$$(x, y) \cdot (u, v) = (xu - \bar{v}y, vx + y\bar{u}), \quad \overline{(x, y)} = (\bar{x}, -y). \quad (\text{A.1})$$

Then $\dim D^M = 2 \dim D$ and $(x, y) \cdot \overline{(x, y)} = (|x|^2 + |y|^2, 0)$.

Proposition A.2 (Mirror chain). *Starting from $\mathbb{R}$ with complex conjugation trivial,*

$$\mathbb{R}^M = \mathbb{C}, \quad \mathbb{C}^M = \mathbb{H}, \quad \mathbb{H}^M = \mathbb{O}, \quad \mathbb{O}^M = \mathbb{S} \text{ (sedenions, not division)}. \quad (\text{A.2})$$

Dimensions 1, 2, 4, 8, 16; only the first four are division algebras (Hurwitz).

Proof. Direct computation of (A.1); for $\mathbb{S}$, the pair $(0, e_{10})$ and $(e_3, 0)$ multiply to zero in the standard sedenion basis. $\square$

A.2 Matrix model

Elements of $J_3(\mathbb{O})$ are

$$X = \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \quad \xi_i \in \mathbb{R}, \quad x_i \in \mathbb{O}, \quad (\text{A.3})$$

with Jordan product $X \circ Y = \frac{1}{2}(XY + YX)$. The trace inner product is $\langle X, Y \rangle = \text{Tr}(X \circ Y)$ and $Q(X) = \text{Tr}(X^2)$.

A.3 Sharp map and rank

The cubic norm is $N(X) = \det X$. The sharp map is

$$X^\# = \text{adj}(X), \quad X^\# \circ X = N(X) E, \quad (X^\#)^\# = N(X) X, \quad (\text{A.4})$$

with $E = \text{diag}(1, 1, 1)$. In diagonal form $X = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$, $X^\# = \text{diag}(\lambda_2\lambda_3, \lambda_3\lambda_1, \lambda_1\lambda_2)$. Thus $\text{rank } X \leq 1 \iff X^\# = 0$.

A.4 Freudenthal identity

$$(X \circ Y)^\# = (N(X)Y + N(Y)X - X^\# \circ Y^\#), \quad (\text{A.5})$$

characterizing $J_3(\mathbb{O})$. Any E_6 -invariant quartic vanishing on $\{X^\# = 0\}$ is proportional to $\langle X^\#, X^\# \rangle$.

Acknowledgments

The author thanks prior UCFT/BAMLP development for the validated spine (archived in `old/`), especially the corrected β -function counts and anomaly arithmetic.

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